BETTERDATA · AI BUILDER CUP 2026 · BFSI

CaseProof

The invoice-financing review copilot

Gemini processes the documents. The bank still decides whether to finance.

01

Process

Gemini extracts the pack and quotes the page.

02

Ask once

Every open gap leaves in one request.

03

Decide

A person records the exception.

04

Publish

The next file follows that policy.

TEAM

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The money is waiting on the paperwork

A fundable invoice sits idle because the receipt does not cover it, and the next email has not been written.

The mismatch

Invoice total and delivery evidence disagree, or part of the goods never arrived.

The second ask

The reviewer writes, waits, and writes again. Earnings wait with that email.

No shared file

The business cannot see the block. The bank cannot show how the pack cleared.

Invoice finance earns when the pack is accepted or declined.

Until then the facility is idle capital. A missing receipt is not a credit decision. It is a stalled file.

The expensive part is asking once, recording the answer, and knowing what still blocks the file.

What a faster, cleaner file is worth

Turn the pack in one sitting

The reviewer opens a case that is already read. Gaps leave together. Earnings start when the pack is clean enough to accept or send back.

Hold the book, not the inbox

The same seller invoice on a second case stays on hold. A copied file on one case is counted once. Neither is a financing verdict.

Leave a file the bank can defend

A frozen PDF and JSON. The client notice omits reviewer names, notes, and model traces. Originals stay hashed.

Publish the rule once

Any desk that can write a checklist can run this copilot on its finance documents. Invoice finance today. Another pack when that policy exists.

One copilot. Five people. One file

The same case. Different authority. The film uses Priya and Sam.

Priya

Reviewer

Sees the quote, the checklist, and one request. Records exceptions. Does not publish policy.

Daniel

Supervisor

Publishes policy and extends budget. Same review tools, plus those two gates.

Sam

Vendor

Uploads the invoice and receipts, answers the request, receives the client notice.

Aisha

Viewer

Reads the completed file. Cannot change the case.

Alex

Walkthrough

Combined roles for a one-person demo. The film uses Priya and Sam.

How a case actually runs

1 Submit

Vendor or bank uploads an English text PDF. Bytes are stored once. A SHA-256 is kept.

2 Extract

Page text, at most 20 pages / 40,000 characters, goes to Gemini. The raw PDF does not.

3 Check

Python commits arithmetic, delivery identity, replacements, and a repeated seller invoice.

4 Ask

The agent issues one request for every open gap, from a published template.

5 Decide

A written message, a dispute, or appearance doubt becomes a human task.

6 Freeze

Completion rechecks everything, then writes a bank summary and a client notice.

An ADK agent on Gemini. One wakeup. One case

How it works a file

Wakes on a saved snapshot. Chooses a tool. The conversation is discarded.

Extracts fields with the printed quote. Plans one request for every open gap.

Compares a written message to the invoice or receipt, then opens a human task.

Asks run_checks before it claims a total. try_complete only after every gate.

It cannot

Approve a facility. Change policy. Pick a recipient. Open another case. Treat PDF text as an order. Compute a total. Waive an unmet rule.

Nine tools. Same validation as the API. None of them approve a facility

read_case · inspect_source · compare_statements · run_checks · prepare_request · issue_request · wait_for_event · open_human_task · try_complete

What Gemini may flag. What the bank can trust

Duplication — certain

Same bytes on one case are hashed and counted once. The same seller, number, and amount on another case is a book hold. Code, not Gemini. Not a finance verdict.

Appearance / forging — a signal

Covered text, cut edges, and a forced template go to a person. Gemini must not claim fraud or authenticity. A careful fake can pass. A legitimate edit can be flagged.

Hidden or malicious prompt — ignored

Text in the PDF or a message is evidence, never an instruction. Orders to complete the case, change rules, or send data are shown to the reviewer. Eval injection flag rate: 1.0.

Quotes — shown only if on the page

Gemini must copy the printed value. An unverified quote is not rendered as evidence. That is how the file stays trustworthy when the model is wrong.

Earlier signal — in context, not retrained

Dismissed and flagged packs stay on the case. Gemini sees them before the next review starts. Weights do not change. A person still records the exception.

The trust rule

Reliability is the wakeup and the lease. Certainty is Python. Safeguarding is a person. Gemini proposes. Code commits. The bank decides.

Policy is the desk's operating manual

Daniel publishes it. Priya cannot. The case pins that version. Gemini may assess extra wording. It cannot waive a core check.

Checklist and holds

Required documents, arithmetic, delivery identity, and the repeated-invoice hold. A text criterion cannot override those.

Templates and timers

The wording of the one request, reminders, and how long the file may wait. The agent may only use a published template.

What the agent may

Request evidence. Open a human task. The page also lists what it may not: finance, pay, change policy, pick a recipient.

Forced template

If the bank wants a letterhead, a hashed reference PDF is required. Later uploads are compared to that layout.

Extra criteria

Added lines are assessed by Gemini against the current pages, with validated quotes. Uncertain findings block completion.

Off, shadow, autonomous

Off stops the agent. Shadow records proposals. Autonomous permits validated commands. Budget or a fault opens a task.

The path a request actually takes

Vendor / bank	API	PostgreSQL	Outbox	Worker	ADK + Gemini
Next.js	FastAPI	Case, jobs, RLS	Dispatcher	Lease	Vertex

Waiting is data

A request, a task, or a timer. No model call is left hanging.

One lease

Another worker cannot apply the same job. Stale revisions return HTTP 412.

One transaction

State, history, and the outbox commit together.

Same product locally

Compose and Cloud Run show the same access-code path and the same case.

What the evaluation actually measured

0

incorrect completions

Agent and rules baseline. 60 cases, 12 families. 12 Oct 2026.

20 / 20

automatic completions

When the pack was already clean, the agent finished without a person.

1.0

injection flag rate

Prompt-in-the-PDF cases were held for a person.

$0.83

model spend

342 Gemini 3.8 Flash calls on Vertex. Mean 11.5 seconds a case.



Staff minutes are a time model, not a measured saving. Zero failures here is not a bank-grade error rate.

Security that ships with the demo

Access code, not IAP

The public web asks for a code. Compose and Cloud Run use the same gate. Judges do not need a Google account.

Private API

Only the web is public. The API, worker, database, and bucket stay private. Web authenticates to the API with its service identity.

Session, CSRF, revision

HttpOnly cookies. CSRF on mutations. Idempotency keys. A stale revision returns 412.

Row security

Runtime roles do not bypass RLS. Organisation context is set for every database access.

Vendor isolation

Sam only sees issued requests and the conversation. Invoice names never grant access.

What Gemini and Search see

Gemini gets page text or rendered pages, not the raw PDF by default. Search gets name, country, and registration — never the invoice.

Built to run beside the bank

Cloud Run

Web public. API and worker private. Same image, ROLE=api or worker.

Cloud SQL 17

The case record. Separate owner, app, and dispatcher roles.

Cloud Storage

Originals and summaries, written once. Public access prevention.

Cloud Tasks

Invokes the worker. Queue capped at 8 concurrent dispatches.

Cloud Scheduler

Reconciles waiting work every minute.

Vertex + ADK

Gemini 3.8 Flash, global. The agent is Google ADK.

Hard bounds, on purpose

Six model calls, sixteen tools, sixty seconds per wakeup. Twelve automatic runs and USD 0.50 a case. Spend is reserved before the call. Non-root containers. Load capacity has not been measured on the hosted project.

What this copilot does not decide

Financing

Does not approve, price, limit, or pay.

Screening

Does not run sanctions, PEP, or adverse-media as a control.

Authenticity

Does not prove a document is genuine, or verify legal identity.

Credit

Research is a sourced brief. It is not a score or continuous watch.

Scope of the PDF

English text, one goods line, SGD, no tax. Three visual pages.

Not in this build

Purchase-order alignment, a fraud-signal engine, and an executive tower remain archived.

Open the prototype

HOSTED URL

https://YOUR-CLOUD-RUN-HOST

TEAM

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The code

On the submission form. Not in this repository.

After the code

A case already read. Quote and checklist on screen.

If you look further

Client notice opens a completed case.

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